ETL Tester Interview Questions

1. What is ETL?

ETL stands for Extract, Transform, and Load. It is a process used to collect data from different source systems, convert it into a required format, and load it into a target database or data warehouse. The transformation step applies business rules and data validation. This process helps organizations analyze and report data effectively.

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2. What is ETL testing?

ETL testing verifies that the data extracted from source systems is correctly transformed and loaded into the target system. The tester checks whether the data is accurate, complete, and consistent. It ensures that transformation rules are correctly applied. ETL testing helps maintain reliable data for business reporting.

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3. What are the stages of ETL?

The ETL process has three main stages: extraction, transformation, and loading. Extraction gathers data from various sources such as databases or files. Transformation applies business rules like filtering, sorting, and calculations. Finally, the transformed data is loaded into the target data warehouse.

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4. What is data extraction?

Data extraction is the first step in the ETL process where data is collected from different source systems. These sources can include databases, applications, flat files, or APIs. The extracted data is usually raw and may contain inconsistencies. This data is then prepared for transformation and further processing.

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5. What is data transformation?

Data transformation is the process of converting raw data into a structured and meaningful format. During transformation, business rules are applied such as filtering records, joining tables, and converting data types. This step ensures data consistency and correctness. It prepares the data for loading into the target system.

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6. What is data loading?

Data loading is the final stage of the ETL process where transformed data is inserted into the target database or data warehouse. This step ensures that the data is stored properly for analysis and reporting. Loading can be performed as a full load or incremental load. The method depends on system requirements.

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7. What is the role of an ETL tester?

An ETL tester validates the data movement from source systems to the target database. They verify data accuracy, completeness, and transformation logic using SQL queries. They also check for issues like duplicates, missing records, and incorrect data formats. Their goal is to ensure reliable data for reporting.

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8. What is a source system?

A source system is the original system from which data is extracted during the ETL process. It may include operational databases, flat files, or external applications. Source systems store the raw data used for business operations. ETL processes collect this data and prepare it for analysis.

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9. What is a target system?

A target system is the destination where transformed data is loaded after processing. Usually, it is a data warehouse or reporting database. The target system stores organized data for analytics and decision-making. It is designed for efficient querying and reporting.

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10. What is source-to-target mapping?

Source-to-target mapping is a document that defines how data fields from the source system correspond to fields in the target system. It also includes transformation rules and data type conversions. ETL testers use this document as a reference to validate data accuracy. It ensures the correct data flow between systems.

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11. What is a data warehouse?

A data warehouse is a centralized repository designed to store large volumes of historical data. It collects data from multiple sources and organizes it for analysis. Unlike operational databases, it is optimized for reporting and business intelligence. Companies use data warehouses to support decision-making.

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12. What is a fact table?

A fact table stores measurable or quantitative data used for analysis, such as sales amounts or transaction counts. It usually contains foreign keys that link to dimension tables. Fact tables represent business events. They are the central tables in data warehouse schemas.

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13. What is a dimension table?

A dimension table contains descriptive information about business entities like customers, products, or locations. It provides context to the data stored in fact tables. Dimension tables help users analyze and categorize business data. They are often connected to fact tables through foreign keys.

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14. What is a star schema?

A star schema is a common data warehouse design where a central fact table connects directly to several dimension tables. The structure resembles a star shape. This design simplifies queries and improves performance. It is widely used in reporting and analytics systems.

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15. What is a snowflake schema?

A snowflake schema is a more normalized version of the star schema. In this structure, dimension tables are divided into additional related tables. This reduces data redundancy but increases complexity. It is used when detailed normalization is required.

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16. What is staging area in ETL?

A staging area is a temporary storage location used during the ETL process. Extracted data is first stored in this area before transformation. It helps in data cleaning, validation, and integration. After processing, the data is moved to the target system.

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17. What is data validation?

Data validation ensures that the data loaded into the target system matches the expected values. Testers verify the data using SQL queries and comparison techniques. This process checks for missing records, incorrect values, and transformation errors. It ensures data accuracy.

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18. What is data reconciliation?

Data reconciliation compares data between the source system and the target system. The purpose is to ensure that both contain the same records and values after ETL processing. This helps detect missing or incorrect data. It is an important step in ETL testing.

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19. What is data completeness testing?

Data completeness testing ensures that all records from the source system are successfully transferred to the target system. Testers compare record counts between source and target tables. If the counts match, it indicates successful data transfer. This prevents data loss during ETL.

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20. What is data accuracy testing?

Data accuracy testing checks whether the values in the target database match the expected results. Testers verify whether transformation rules are correctly applied. This ensures that the data stored in the warehouse is reliable. Accurate data is essential for decision-making.

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21. What is data quality testing?

Data quality testing verifies whether the data is clean, valid, and consistent. It checks for duplicate records, missing values, and incorrect formats. High-quality data ensures accurate reporting. Poor data quality can lead to incorrect business decisions.

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22. What is incremental load?

Incremental load means loading only the new or updated records from the source system into the target system. This approach saves time and reduces system load. It is commonly used for daily or scheduled ETL processes.

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23. What is full load?

Full load means transferring all records from the source system to the target system every time the ETL job runs. It is usually used during the initial data migration. Although simple, it can take longer for large datasets.

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24. What is surrogate key?

A surrogate key is a system-generated unique identifier used in data warehouse tables. It replaces natural keys from source systems. Surrogate keys simplify data management and improve performance. They are commonly used in dimension tables.

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25. What is data granularity?

Data granularity refers to the level of detail stored in the data warehouse. High granularity means very detailed data, while low granularity means summarized data. Choosing the right level helps optimize storage and reporting performance.

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26. Why is SQL important for ETL testers?

SQL helps ETL testers retrieve, compare, and validate data from databases. Testers use SQL queries to verify record counts, data accuracy, and transformation logic. It is the primary tool for data validation. Strong SQL skills are essential for ETL testing.

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27. What is duplicate data testing?

Duplicate data testing checks whether the ETL process loads the same record multiple times. Testers use SQL queries to identify duplicate entries. Preventing duplicates ensures data consistency. Duplicate records can affect reporting accuracy.

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28. What is null value testing?

Null value testing verifies how the ETL process handles missing or empty values. Testers check whether null values are correctly managed according to business rules. Proper handling prevents errors in reports and analysis.

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29. What is transformation testing?

Transformation testing verifies that data transformation rules are correctly applied during the ETL process. This includes calculations, filters, and data conversions. Testers compare transformed data with expected results. It ensures the business logic is correctly implemented.

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30. What is ETL job?

An ETL job is a process or workflow that performs extraction, transformation, and loading tasks. It can be scheduled to run automatically at specific times. ETL jobs move and process data efficiently between systems.

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31. What is batch processing?

Batch processing executes ETL jobs at scheduled intervals instead of processing data in real time. It is commonly used for large data volumes. This method processes data in groups or batches.

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32. What is data profiling?

Data profiling analyzes the structure and quality of source data before starting the ETL process. It helps identify issues such as missing values or inconsistent formats. Understanding the data helps design better transformation rules.

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33. What is data cleansing?

Data cleansing removes incorrect, duplicate, or incomplete data from the dataset. This improves data quality before loading it into the data warehouse. Clean data ensures accurate analysis.

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34. What is regression testing in ETL?

Regression testing ensures that changes made to ETL processes do not affect existing functionalities. Testers re-run previous test cases after modifications. This confirms that the system still works correctly.

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35. What is performance testing in ETL?

Performance testing checks how efficiently the ETL process handles large volumes of data. It measures processing time and system resource usage. Good performance ensures timely data availability.

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36. What is data truncation testing?

Data truncation testing verifies whether data gets cut or shortened during transformation. This can happen if the target column size is smaller than the source data. Testers ensure that no important data is lost.

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37. What is error handling in ETL?

Error handling ensures that errors during the ETL process are captured and logged properly. This helps developers identify and fix issues quickly. Proper error handling improves system reliability.

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38. What is reject record testing?

Reject record testing verifies that invalid or incorrect records are properly rejected during ETL processing. These records are stored in error logs or reject tables. This prevents bad data from entering the warehouse.

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39. What is data integration?

Data integration combines data from multiple sources into a single unified view. ETL processes perform this integration. It helps organizations analyze data from different systems together.

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40. What is data migration?

Data migration is the process of transferring data from one system or database to another. It may happen during system upgrades or platform changes. ETL tools often assist in migration.

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41. What are ETL tools?

ETL tools automate the process of extracting, transforming, and loading data. Examples include Informatica, Talend, DataStage, and SSIS. These tools simplify data integration and improve efficiency.

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42. What is Slowly Changing Dimension (SCD)?

SCD refers to managing changes in dimension data over time. It tracks historical changes in dimension records. This helps maintain accurate historical reporting.

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43. What are types of SCD?

Common SCD types include Type 1, Type 2, and Type 3. Type 1 overwrites old data, Type 2 keeps historical records, and Type 3 stores limited history. Each type is used depending on business requirements.

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44. What is data comparison testing?

Data comparison testing compares records between source and target databases. Testers ensure that values match after ETL processing. This confirms correct data transfer.

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45. What is ETL workflow?

An ETL workflow defines the sequence of ETL jobs and tasks. It controls how data flows from source to target. Workflows ensure the correct execution order.

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46. What is test case in ETL testing?

A test case describes a specific condition used to validate ETL functionality. It includes input data, expected results, and validation steps. Test cases help ensure systematic testing.

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47. What is test scenario?

A test scenario is a high-level description of a testing activity. It defines what functionality needs to be tested. Test scenarios help organize testing efforts.

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48. What challenges do ETL testers face?

ETL testers often deal with large data volumes and complex transformation rules. They also face data quality issues and system performance challenges. Proper testing strategies help manage these problems.

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49. Why is ETL testing important?

ETL testing ensures that the data used for reporting and analysis is accurate and reliable. Without proper testing, incorrect data may enter the data warehouse. This can lead to wrong business decisions.

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50. What skills are required for an ETL tester?

An ETL tester should have strong SQL skills, understanding of data warehousing concepts, and knowledge of ETL tools. They must also understand testing methodologies and data validation techniques.
